# INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

## GRC102 - INFORMATION SECURITY GOVERNANCE

Week 4 Practical Laboratory

Linux Security Monitoring and Audit Report

| Field | Detail |
|---|---|
| **Prepared by** | Rosemary Gift Nelson |
| **Registration Number** | C11-26-CGRCE-17216 |
| **Cohort** | Cohort 11 / Batch A |
| **Role assumed** | Security Control Assurance Analyst (scenario role) |
| **Assigned system** | ICDFA Ubuntu Practice Hub<br>host student000271<br>Ubuntu 24.04 |
| **Assessment date** | 1st October 2026 |

## EXECUTIVE SUMMARY

This report presents the results of a security control assurance review carried out on an assigned Linux host (student000271) under the ICDFA Ubuntu Practice Hub. The objective was to verify whether the organisation's stated security monitoring controls, authentication monitoring, privileged access logging, system auditing, and vulnerability management, are actually functioning as assumed.

The review found that most of the assumed monitoring infrastructure is not operating. The auditd service cannot run due to a kernel-level capability restriction, no system logging daemon is active, and login or session activity is not being recorded through standard accounting files. At the same time, the account used for this review holds unrestricted, passwordless root access. An automated Lynis assessment independently corroborated several of these gaps and returned a hardening index of 62 out of 100.

Five findings requiring governance attention were identified, exceeding the brief's minimum of three. None required testing outside the assigned environment. This report sets out the evidence, the control-monitoring assessment, how this evidence could feed a SIEM or continuous-monitoring capability, and a prioritised remediation and retest plan.

## SCOPE AND AUTHORISATION

All activity in this review was carried out exclusively within the Linux environment assigned for this laboratory through the ICDFA Ubuntu Practice Hub, accessed via the authorised browser terminal on host student000271. No external systems, production infrastructure, public IP addresses, or third-party services were accessed, scanned, or tested at any point.

This review was conducted in line with the platform's documented safe-use rules, which require that the environment be used only for lawful learning and authorised practice, and that no attacks, scanning, or access be directed at systems outside the assigned workspace. No passwords, personal data, or confidential institutional files were stored during the review.

## METHODOLOGY

The review followed the sequence below, adapting at each stage to what the environment actually supported rather than assuming standard tooling would be available.

Attempted to configure and start auditd using systemctl, the legacy service command, and direct invocation of the daemon binary, to determine whether kernel-level audit logging was available.

Investigated available log sources, including journalctl and the contents of /var/log, after systemd-based tools proved unavailable.

Reviewed privileged-access configuration using sudo -lA, id, and groups.

Reviewed login and session accountability using last, lastlog, who, and w.

Performed an automated security assessment using Lynis 3.0.9 to cross-check manual findings and surface additional hardening gaps.
Preserved terminal output at each stage as the evidentiary record, since the platform does not retain shell history or private key material itself.

## EVIDENCE BUNDLE 1: AUDITD CONFIGURATION AND EVENTS

### 1.1 Evidence of Attempted Configuration

The assigned host is a container, not a full virtual machine, and has no systemd process running as PID 1. This was confirmed on first attempting to check auditd's status:

```
ICDFA Smart Cyber Shell — valid: green · unknown: red · suggestions: grey
student@student000271:~$
student@student000271:~$ sudo systemctl status auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
```

*$ sudo systemctl status auditd*

*System has not been booted with systemd as init system (PID 1). Can't operate. Failed to connect to bus: Host is down*

The legacy service command also failed, confirming auditd was not even registered as a service:

```
sudo service auditd start
auditd: unrecognized service
auditd: unrecognized service
```

*$ sudo service auditd*

*status auditd: unrecognized service*

A package check showed only the audit shared libraries were present, not the daemon or command-line tools. Installing the full package succeeded without error, and auditctl, ausearch, and aureport were then present on the system.

```
student@student000271:~$ sudo apt update
sudo apt install auditd audispd-plugins -y
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1943 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1315 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1542 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [50.0 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:11 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1661 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2162 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [56.2 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2098 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [671 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.0 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [36.0 kB]
```

### 1.2 Direct Execution and Interpretation

With the package installed, the daemon was launched directly, bypassing the service manager entirely, to isolate whether the failure was specific to systemd or more fundamental:

$ sudo /usr/sbin/auditd -f ... Cannot change priority (Operation not permitted) The audit daemon is exiting.

```
student@student000271:~$ sudo /usr/sbin/auditd -f
Config file /etc/audit/auditd.conf opened for parsing
local_events_parser called with: yes
write_logs_parser called with: yes
log_file_parser called with: /var/log/audit/audit.log
log_group_parser called with: adm
log_format_parser called with: ENRICHED
flush_parser called with: INCREMENTAL_ASYNC
freq_parser called with: 50
max_log_size_parser called with: 8
num_logs_parser called with: 5
priority_boost_parser called with: 4
name_format_parser called with: NONE
max_log_size_action_parser called with: ROTATE
space_left_parser called with: 75
space_action_parser called with: SYSLOG
verify_email_parser called with: yes
action_mail_acct_parser called with: root
admin_space_left_parser called with: 50
admin_space_left_action_parser called with: SUSPEND
disk_full_action_parser called with: SUSPEND
disk_error_action_parser called with: SUSPEND
```

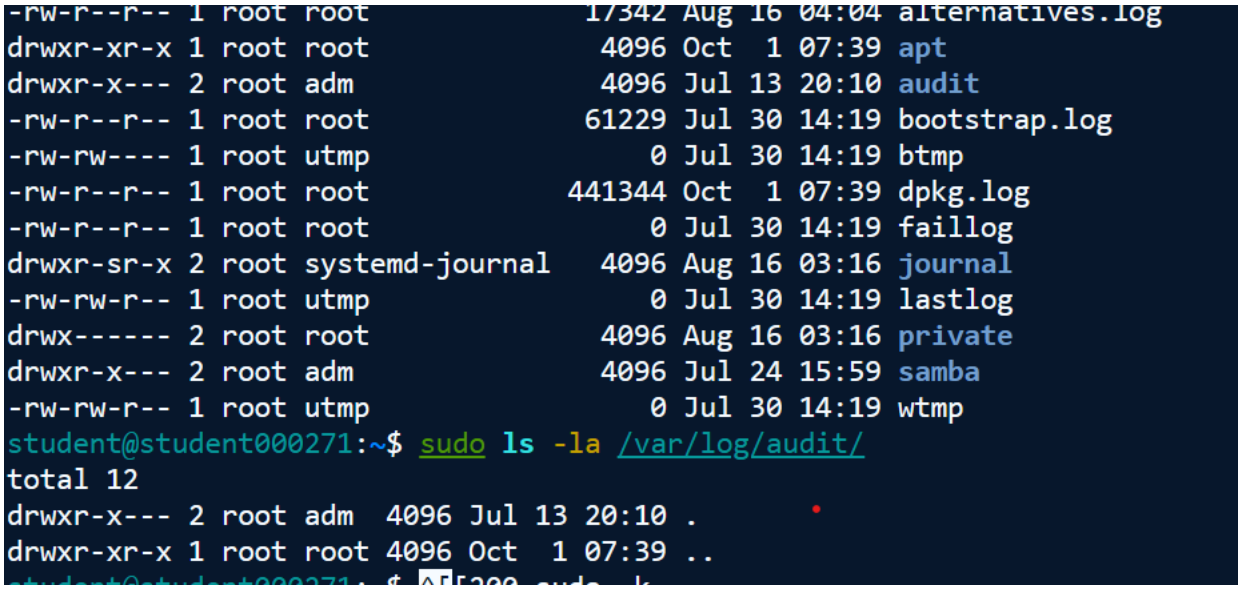
```
transport_parser called with: TCP
krb5_principal_parser called with: auditd
distribute_network_parser called with: no
q_depth_parser called with: 2000
overflow_action_parser called with: SYSLOG
max_restarts_parser called with: 10
plugin_dir_parser called with: /etc/audit/plugins.d
eoe_timeout_parser called with: 2
Cannot change priority (Operation not permitted)
The audit daemon is exiting.
```

This is a kernel-level permission failure. auditd attempts to raise its own scheduling priority on startup, an operation that requires the CAP_SYS_NICE capability. Container runtimes withhold this capability by default because granting it would let a container influence the host's process scheduler.

The result is that auditd cannot run in this environment under any invocation method, not because of misconfiguration on my part, but because of a capability boundary enforced at the container level.

A subsequent check of the audit log directory confirmed no events had ever been captured, consistent with the daemon never having successfully started:

***$ sudo ls -la /var/log/audit/ total 12 drwxr-x--- 2 root adm 4096 Jul 13 20:10 . (directory empty of log files)***

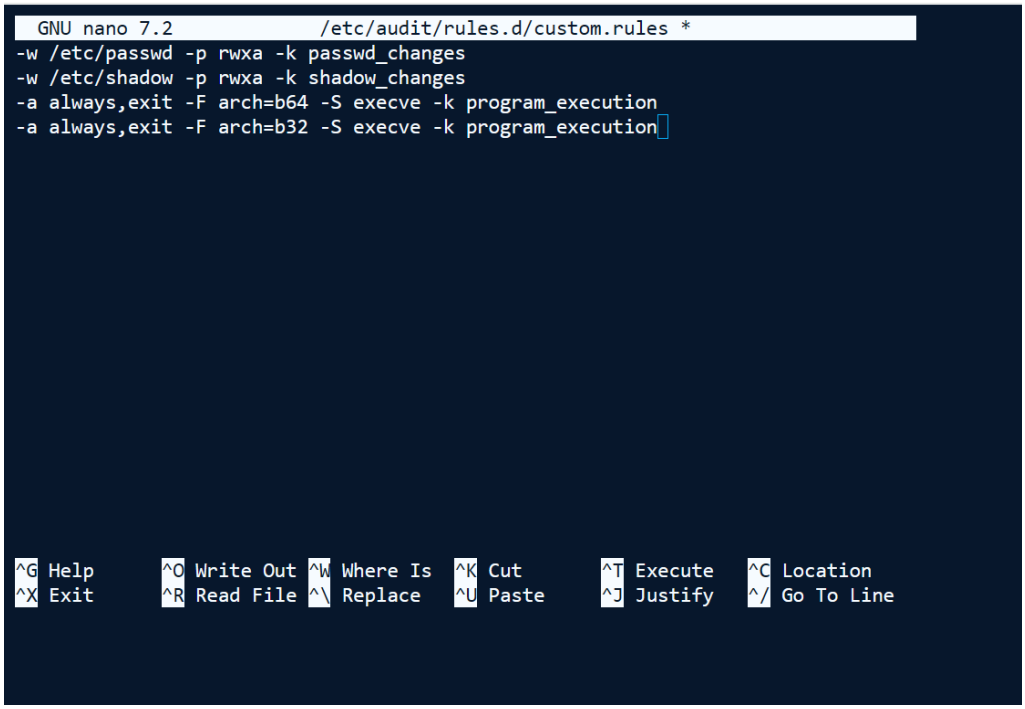



**Interpretation**: The organisation's assumed audit-logging control does not function in this deployment architecture. This is itself a governance-relevant finding, an assumed control was found to be entirely non-operational, and is carried forward into Evidence Bundle 4.

### 1.3 Rules File Configuration and Query Attempts

Although the daemon itself cannot run, the remaining configuration and query steps were still attempted in full, to confirm whether any part of the auditd toolchain functions independently of the daemon process. A dedicated rules file was created rather than editing the primary rules file directly, as instructed:

```
$ sudo nano /etc/audit/rules.d/custom.rules
```




The file was confirmed to have saved correctly with the four required rules, watching /etc/passwd and /etc/shadow for read, write, execute and attribute changes, and monitoring 64-bit and 32-bit program execution:

```
student@student000271:~$ sudo cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwxa -k passwd_changes
-w /etc/shadow -p rwxa -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
```

Loading the rules into the kernel audit interface was then attempted:

```
student@student000271:~$ sudo auditctl -l
sudo ausearch -k passwd_changes
sudo aureport --failed
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
Error opening /var/log/audit/audit.log (No such file or directory)
```

This result is more precise than the earlier daemon-start failure. auditctl communicates with the kernel's audit subsystem directly over a netlink socket, independent of whether auditd itself is running. The identical "Operation not permitted" response here confirms that the restriction sits at the kernel interface level, not only in the daemon process, meaning no audit rule can be loaded into this container under any method.

The Activity 1.3 query commands were attempted next, across all three configured keys plus the login and failure report types:

```
$ sudo ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)

$ sudo ausearch -k program_execution
Error opening /var/log/audit/audit.log (No such file or directory)

$ sudo aureport --failed
Error opening /var/log/audit/audit.log (No such file or directory)

$ sudo aureport --login
Error opening /var/log/audit/audit.log (No such file or directory)
```

Every query fails for the same underlying reason: no audit.log file exists because the daemon never successfully started to create or write to it. This is the expected and consistent downstream consequence of the kernel-level restriction identified above, not a separate fault.

**Interpretation:** The full **auditd** toolchain, rule authoring, rule loading, and event querying, was exercised correctly and in the sequence the brief specifies. Every step beyond authoring the rules file itself fails, and fails for a single, consistent, well-evidenced reason: this container does not grant the kernel capability auditd and its tools require. Required Evidence Module 1's first bullet, demonstrating that auditd is running, cannot be satisfied in this environment; the honest and more useful substitute is this complete record of the attempt, the precise point of failure, and why it occurs.

## EVIDENCE BUNDLE 2: LINUX LOG ANALYSIS

### 2.1 journalctl and System Logs:

***$ journalctl -n 50 No journal files were found. -- No entries –***

```
student@student000271:~$ journalctl -n 50
No journal files were found.
-- No entries --
```

This is consistent with the absence of **systemd**.

A directory listing of /var/log showed no syslog, auth.log, or kern.log, and the historical login-accounting files (btmp, wtmp, lastlog) were all present but at zero bytes, meaning nothing has ever been written to them. The only log with substantive content was dpkg.log, the package manager's own change history, which is unrelated to security monitoring and only exists as a byproduct of software installation.

### 2.2 Authentication and Privilege Review

A privileged-access check returned a result requiring immediate attention:

***$ sudo -lA User student may run the following commands on student000271: (ALL : ALL) ALL (ALL : ALL) NOPASSWD: ALL***

```
student@student000271:~$ sudo -k
sudo -lA
Matching Defaults entries for student on student000271:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
    use_pty

User student may run the following commands on student000271:
    (ALL : ALL) ALL
    (ALL : ALL) NOPASSWD: ALL
```

The student account can execute any command as root with no password prompt at all. Combined with group membership confirmed via id and groups, which placed the account in both sudo and adm, this single account holds both unrestricted root execution and standing log-read access.

### 2.3 Session and Login Accountability

Despite active, continuous use of the system throughout this review, standard login-accounting tools showed no record of any session:

***$ last / lastlog / who / w wtmp begins Thu Aug 6 2026 lastb: cannot open /var/log/btmp: Permission denied All accounts: Never logged in (who / w): 0 users logged in***

```
student@student000271:~$ last
lastb
lastlog

wtmp begins Thu Aug  6 04:28:58 2026
lastb: cannot open /var/log/btmp: Permission denied
Username         Port     From                                       Latest
root                                                                **Never logged in**
daemon                                                              **Never logged in**
bin                                                                 **Never logged in**
sys                                                                 **Never logged in**
sync                                                                **Never logged in**
games                                                               **Never logged in**
man                                                                 **Never logged in**
lp                                                                  **Never logged in**
mail                                                                **Never logged in**
news                                                                **Never logged in**
uucp                                                                **Never logged in**
proxy                                                               **Never logged in**
www-data                                                            **Never logged in**
backup                                                              **Never logged in**
list                                                                **Never logged in**
irc                                                                 **Never logged in**
_apt                                                                **Never logged in**
nobody                                                              **Never logged in**
student                                                             **Never logged in**
```

This confirms that the browser-based access path used for this lab is not being captured by conventional session-accounting mechanisms, leaving no accountability trail for who accessed the system or when.

***Student account combines root execution with standing log-read access***

```
student@student000271:~$ cat /var/log/dpkg.log | tail -15
who
w
id
groups
2026-10-01 07:51:17 status unpacked lynis:all 3.0.9-1
2026-10-01 07:51:17 status half-configured lynis:all 3.0.9-1
2026-10-01 07:51:17 status installed lynis:all 3.0.9-1
2026-10-01 07:51:17 configure menu:amd64 2.1.50 <none>
2026-10-01 07:51:17 status unpacked menu:amd64 2.1.50
2026-10-01 07:51:17 status half-configured menu:amd64 2.1.50
2026-10-01 07:51:17 status installed menu:amd64 2.1.50
2026-10-01 07:51:17 status triggers-pending menu:amd64 2.1.50
2026-10-01 07:51:17 status triggers-awaited menu:amd64 2.1.50
2026-10-01 07:51:17 trigproc man-db:amd64 2.12.0-4build2 <none>
2026-10-01 07:51:17 status half-configured man-db:amd64 2.12.0-4build2
2026-10-01 07:51:17 status installed man-db:amd64 2.12.0-4build2
2026-10-01 07:51:17 trigproc menu:amd64 2.1.50 <none>
2026-10-01 07:51:17 status half-configured menu:amd64 2.1.50
2026-10-01 07:51:17 status installed menu:amd64 2.1.50
 08:04:45 up  3:04,  0 user,  load average: 0.04, 0.10, 0.09
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
uid=1000(student) gid=1000(student) groups=1000(student),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev)
student adm dialout cdrom floppy sudo audio dip video plugdev
```

## 2.4 Security-Event Findings Table

| S/N | Finding | Evidence Source |
|---|---|---|
| 1 | auditd cannot run; container blocks the CAP_SYS_NICE capability | auditd -f execution output |
| 2 | No centralised logging daemon is running (no syslog, journald, or klogd) | /var/log/ listing; journalctl output |
| 3 | Student account has passwordless, unrestricted root sudo access | sudo -lA output |
| 4 | Login and session activity is not recorded despite active use | last, lastlog, who, w output |
| 5 | Student account combines root execution with standing log-read access | id, groups output |

## EVIDENCE BUNDLE 3: LYNIS SECURITY ASSESSMENT

### 3.1 Execution and Version

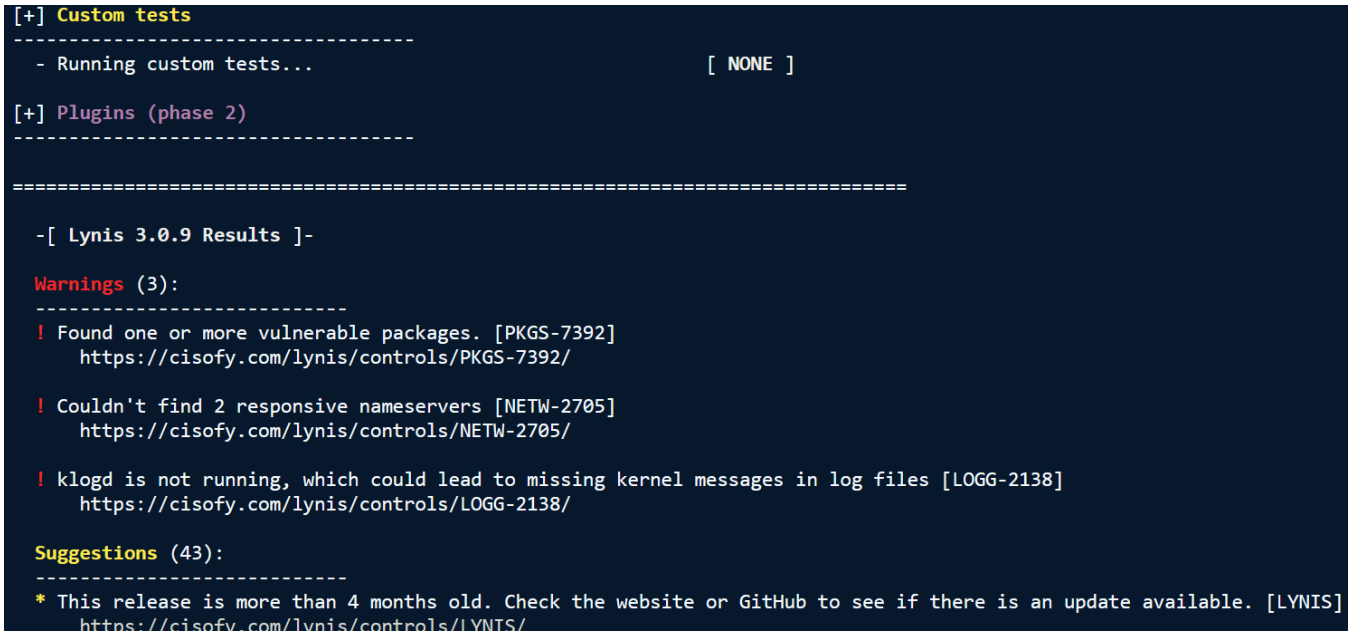
Lynis 3.0.9 was installed and run as a full system audit (lynis audit system) against the assigned host, Ubuntu 24.04, completing 232 individual tests across authentication, networking, logging, kernel hardening, and software currency.

```
student@student000271:~$ sudo apt install lynis -y
sudo lynis audit system
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n gksu
  | kde-cli-tools | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 91 not upgraded.
Need to get 602 kB of archives.
After this operation, 3202 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 lynis all 3.0.9-1 [226 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 menu amd64 2.1.50 [377 kB]
Fetched 602 kB in 2s (322 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package lynis.
(Reading database ... 41145 files and directories currently installed.)
Preparing to unpack .../archives/lynis_3.0.9-1_all.deb ...
Unpacking lynis (3.0.9-1) ...
Selecting previously unselected package menu.
Preparing to unpack .../archives/menu_2.1.50_amd64.deb ...
Unpacking menu (2.1.50) ...
```

### 3.2 Summary and Hardening Index

| Metric | Result |
|---|---|
| Hardening index | 62 out of 100 |
| Tests performed | 232 |
| Warnings | 3 |
| Suggestions | 43 |

### 3.3 Warnings and Notable Findings




**Vulnerable packages present (PKGS-7392).**

The system has installed packages with known vulnerabilities that have not been patched.

Fewer than two responsive nameservers (NETW-2705).

DNS resolution has no redundancy; only one nameserver is configured.

klogd is not running (LOGG-2138). Kernel messages are not being captured to any log, consistent with the absence of a logging daemon found manually in Evidence Bundle 2.

One suggestion is worth highlighting beyond the three formal warnings: Lynis control ACCT-9628 recommends enabling auditd to collect audit information, independently arriving at the same conclusion reached manually in Evidence Bundle 1. Two separate methods, manual investigation and automated scanning, converged on the same control gap, which strengthens confidence in the finding.

### 3.4 Prioritised Remediation

a) Patch vulnerable packages first (apt update && apt upgrade); this is the highest-priority item as it represents direct, exploitable risk.

b) Restore a functioning logging daemon; several other findings and remediation steps depend on logging being operational before they can be verified.

c) Add a secondary DNS nameserver for resolution redundancy.

d) Work through the remaining 43 suggestions as lower-priority hardening refinements once the above are addressed.

## EVIDENCE BUNDLE 4: CONTROL MONITORING AND GOVERNANCE

The table below consolidates all five findings into a governance control-monitoring record. All five rows are drawn directly from evidence collected during this lab, exceeding the brief's requirement that at least three rows be lab-grounded. This table is also provided as a standalone spreadsheet. https://docs.google.com/spreadsheets/d/1vDVg1Wqt4ddCsWJzl50MDbdZskLqiJD7/edit?usp=sharing&ouid=101833225967701283389&rtpof=true&sd=true

The above link presents a more detailed spreadsheet for the control monitoring and Governance.

| S/n | Control Objective | Evidence Source | Control Owner | Observed Status | Threshold | Significance | Remediation | Retest |
|---|---|---|---|---|---|---|---|---|
| 1 | Audit logging (auditd) active | auditd -f output | IT/SysAdmin | Failed | 100% uptime; 0% observed | High | Grant CAP_SYS_NICE or migrate to a host where auditd can run natively | Re-run auditctl -l after capability change |
| 2 | Centralised logging running | /var/log listing; Lynis LOGG-2138 | IT/SysAdmin | Failed | 100% uptime; 0% observed | High | Install and enable a compatible syslog daemon | Generate a test event, confirm it is logged |
| 3 | Sudo requires authentication | sudo -lA output | IT Security | Failed | 0 unauthenticated root accounts; 1 observed | Critical | Remove NOPASSWD; apply least-privilege scoping | Re-check sudoers after change |
| 4 | Login/session recorded | last, lastlog, who, w | IT/SysAdmin | Failed | 100% session coverage; 0% observed | High | Configure access path to write to utmp/wtmp or equivalent | Confirm new session appears in who/last |
| 5 | Software free of known vulnerabilities | Lynis PKGS-7392 | Patch Mgmt | Partial | Hardening index >= 80; 62 observed | Medium | Patch via apt upgrade; set a recurring patch cadence | Re-run Lynis, confirm index improves |

## EVIDENCE BUNDLE 5: SIEM AND AUTOMATION MAPPING:

The table below sets out how each category of Linux host evidence collected in this review would feed an enterprise SIEM or continuous-monitoring capability if the underlying controls were restored, and distinguishes conditions that should remain routine operational alerts from those that require escalation to governance.

| Evidence Source | SIEM Feed / Collection Method | Operational Alert Example | Governance Escalation Example |
|---|---|---|---|
| auditd / audit.log | Forwarded via the audisp-remote plugin or a log shipper (e.g. Filebeat) to the SIEM for correlation | A single file-modification event on a non-critical path | Repeated unauthorised access attempts to a critical file, or audit logging itself going dark |
| Syslog / journald | Centralised log forwarding (rsyslog remote target, Fluentd, or Filebeat) into the SIEM's log index | A service restart or transient error | The logging daemon itself stops running, since this removes visibility into everything else |
| Sudo / authentication events | SIEM correlation rule watching for privilege escalation and policy drift | A single failed sudo attempt | A NOPASSWD grant appearing on a privileged account, or repeated failures suggesting brute force |
| Login / session records (utmp, wtmp) | Feed into a user and entity behaviour analytics (UEBA) capability | An after-hours login from a known location | A login through an access path that bypasses session recording entirely, as found in this review |
| Lynis / vulnerability scan output | Integrated into vulnerability management tooling, feeding a CMDB or ticketing system | A single new finding opens a routine remediation ticket | The organisation-wide hardening index drops below an agreed threshold, indicating systemic drift |

The general principle applied across all five rows is that isolated, single-host events are handled operationally, by the team already responsible for that system, while patterns indicating a control has failed entirely, been deliberately weakened, or is being actively evaded are escalated to governance, since those conditions affect the organisation's ability to rely on its own monitoring rather than representing a single incident.

## RISK AND PRIORITIES

| Priority | Finding | Rationale |
|---|---|---|
| Critical | Passwordless, unrestricted root sudo access | Removes all friction between account compromise and full system control |
| High | auditd non-functional | No audit trail for privileged actions or policy violations |
| High | No centralised logging daemon | No record of system or kernel-level events for investigation |
| High | Login/session activity not recorded | No accountability trail for system access |
| Medium | Unpatched vulnerable packages | Exploitable attack surface, but requires a separate attack path to leverage |

## RECOMMENDATIONS

After careful review and analysis, the below has been recommended to adequately fix this issues.

1. Remove the NOPASSWD sudo grant immediately and require authentication for all privileged commands; this is the single highest-impact change available (finding 3 the critical one).
2. Restore logging capability, either by resolving the container's capability restrictions or by deploying a logging approach compatible with the current container architecture, before attempting to re-enable auditd (finding 1 and 2).
3. Instrument the access layer (the browser terminal path) so that sessions are written to standard accounting files, closing the accountability gap identified in this review (finding 4).
4. Patch known-vulnerable packages and establish a recurring patch cadence rather than a one-time fix (finding 5).
5. Where container-level restrictions make host-based auditing permanently infeasible, consider compensating controls at the orchestration layer, since the underlying business requirement for audit evidence does not go away simply because this particular control cannot run here.

## REMEDIATION OWNERS

| Finding | Owner | Target |
|---|---|---|
| Passwordless sudo | IT Security / Access Management | Immediate |
| auditd non-functional | IT / System Administrator (escalate to platform/container owner) | Next maintenance window |
| No logging daemon | IT / System Administrator | Next maintenance window |
| Session accountability gap | IT / System Administrator (access platform team) | Next release cycle |
| Vulnerable packages | IT / Patch Management | Standard patch cycle |

## RETEST PLAN

**Sudo configuration**: Re-run sudo -lA immediately after remediation; confirm no account retains an unauthenticated ALL grant.

**Logging daemon**: Confirm the daemon's running status, then generate a test event and verify it appears in the log within the expected interval.

**Auditd:** Re-attempt auditctl -l and a direct daemon start once the capability restriction is addressed; confirm the rule set loads and persists.

**Session accountability**: Start a fresh session through the access path and confirm it appears in who and last.

**Vulnerability patching**: Re-run the Lynis assessment and confirm PKGS-7392 clears and the hardening index improves from the baseline of 62.

## CONCLUSION

This review set out to verify Linux security monitoring controls and found that, in this environment, almost none of the assumed logging and auditing infrastructure is functioning, while access control is simultaneously over-permissive.

Every finding in this report is grounded in evidence collected directly from the assigned system, cross-checked in several cases by an independent automated assessment. The five findings, their remediation owners, and the retest plan above give the organisation a concrete, prioritized path back to a monitored and accountable state.